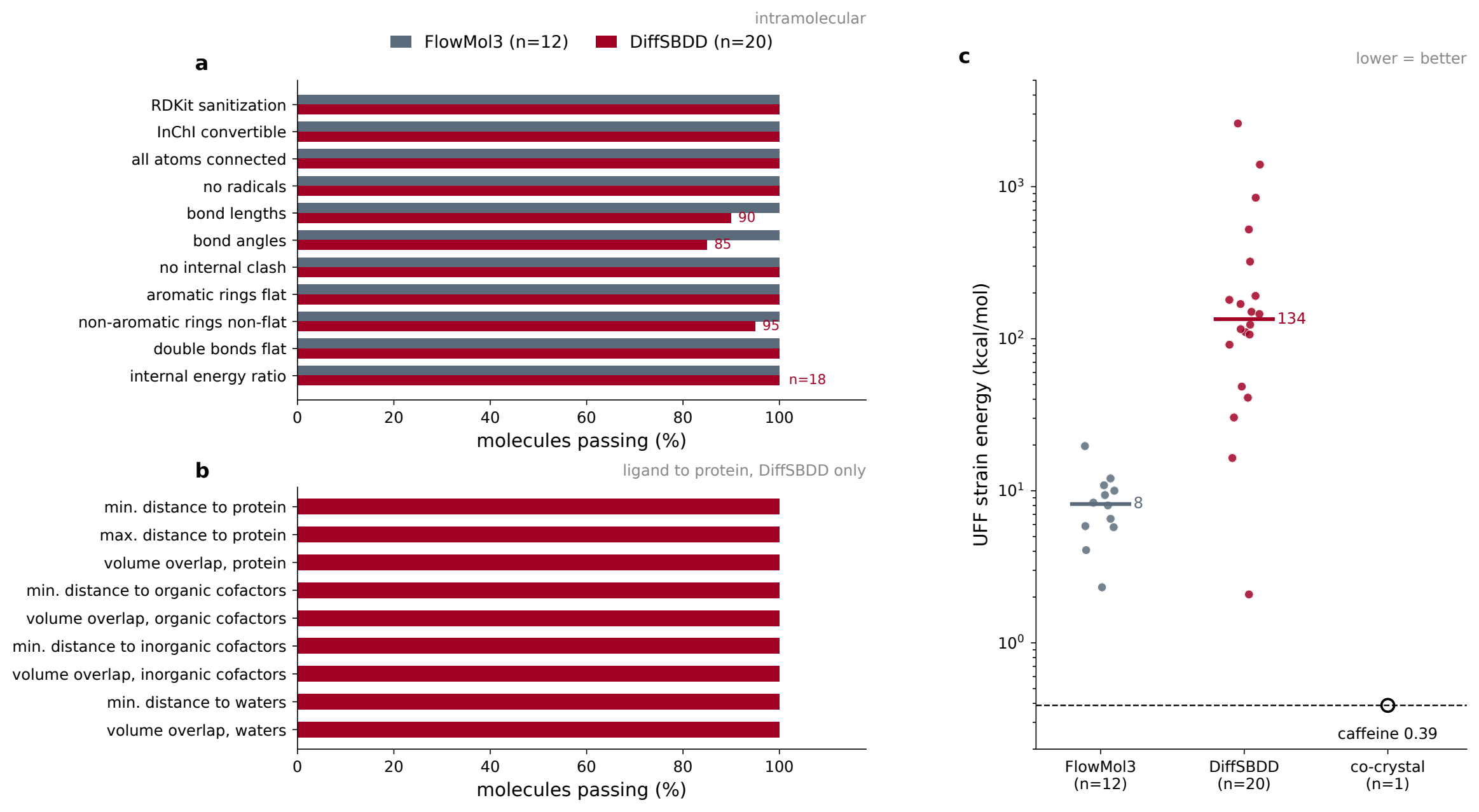
intramolecular
FlowMol3 (n=12)
DiffSBDD (n=20)
a
RDKit sanitization
InChI convertible
all atoms connected
no radicals
bond lengths
bond angles
no internal clash
aromatic rings flat
non-aromatic rings non-flat
double bonds flat
internal energy ratio
90
85
95
n=18
0
20
40
60
80
100
molecules passing (%)
b
ligand to protein, DiffSBDD only
min. distance to protein
max. distance to protein
volume overlap, protein
min. distance to organic cofactors
volume overlap, organic cofactors
min. distance to inorganic cofactors
volume overlap, inorganic cofactors
min. distance to waters
volume overlap, waters
0
20
40
60
80
100
molecules passing (%)
c
lower = better
UFF strain energy (kcal/mol)
10^3
10^2
10^1
10^0
8
134
caffeine 0.39
FlowMol3
(n=12)
DiffSBDD
(n=20)
co-crystal
(n=1)